DUBSAUCE

Dub Siren — Daisy Seed3 Desktop Dev Kit · Player's Manual · Build v0.4.1

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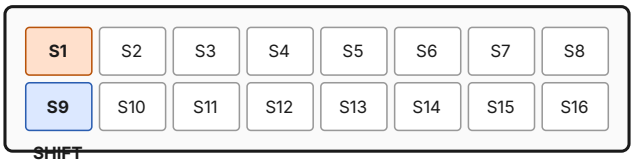
1. Overview

DubSauce is a single-voice dub siren built for the Daisy Seed3 Desktop Dev Kit. It centers on a classic analog-style siren signal path — oscillator → filter → VCA — driven by a deep LFO/wail engine in the spirit of the BeniDub DS71 and SCAT-style units, with a built-in tape echo and reverb send/return bus.

Eight knobs (VR1–VR8) are reused across four pages, selected with the SHIFT pad, so every parameter of the voice and the FX stays reachable without crowding the panel.

2. Layout & Controls

The 16 pads are arranged as two rows of eight. S1 (top-left) is the trigger pad. S9 (bottom-left) is SHIFT, used to step through the four pot pages.



Control	Function
S1	Trigger pad — fires the siren per SW18's mode. LED lit while sounding.
S9 (SHIFT)	Advances the pot page: LIVE → SHAPE → DEEP → FX → LIVE … Bottom-row LEDs light up left-to-right to show the current page count (1–4).
SW18 (3-pos toggle)	**Up** — momentary (sounds while held). **Center** — mute. **Down** — latch (each press toggles on/off).
GATE IN 1	External trigger input — same behavior as S1.
CV OUT 1	Mirrors the siren's internal wail/LFO signal, for patching into other gear.
MIDI IN	Note-On/Off plays and gates the siren at the MIDI pitch.
LINE IN L/R	Routed into the FX bus alongside the siren — see Section 8.

3. Playing the Siren

Set SW18 to **Up** for classic hands-on dub siren playing — press and hold S1 (or send a gate on GATE IN 1) to sound the siren, release to stop. Flip to **Down** to latch it on/off with each press — useful for hands-free drones. **Center** is silence regardless of what's held.

Pitch defaults to the Tune knob (LIVE page, VR1). Sending a MIDI note overrides Tune with the note's pitch for as long as that note is held.

4. Pot Pages — LIVE PAGE 1

The performance page — the knobs you'll ride most while playing live.

Knob	Parameter	Range / Notes
VR1	Tune	30 Hz – 9 kHz (DS71-style exponential range)
VR2	Wail Rate	0.1 – 15 Hz — speed of the pitch LFO
VR3	Wail Depth	0 – 3 octaves of LFO pitch swing
VR4	Sweep	±2 octaves — one-shot pitch sweep amount
VR5	Decay	20 ms – 4 s — tail length after release
VR6	Tone	120 Hz – 18 kHz — filter cutoff
VR7	Delay Send	0–1 — siren level sent to the tape echo bus
VR8	Reverb Send	0–1 — siren level sent to the reverb bus

5. Pot Pages — SHAPE PAGE 2

Shapes the raw tone of the oscillator and filter — think of this as the siren's "voice."

Knob	Parameter	Range / Notes
VR1	Osc Waveform	6 steps: Sine · Pulse · Saw · Tri · Blend · RingFM
VR2	PW / Blend	Pulse width (square waves) or blend amount, depending on waveform
VR3	Sub Level	0–1 — sub-oscillator one octave down
VR4	Drive	0–1 — pre-filter saturation
VR5	Noise Level	0–1 — white noise blended into the signal
VR6	LFO Waveform	8 steps: Sine · Tri · Square · Ramp Up · Ramp Down · Sample&Hold · Exp Env · Log Env
VR7	Swing	0–0.75 — LFO rhythmic swing/shuffle
VR8	Resonance	0–0.9 (hard-capped for stability) — filter resonance

> Resonance is deliberately capped below self-oscillation. If the filter ever sounds like it's losing signal at extreme settings, the firmware has a built-in safety reset — just back the knob off.

6. Pot Pages — DEEP PAGE 3

Deeper modulation and envelope shaping — set-and-forget territory for sound design.

Knob	Parameter	Range / Notes
VR1	Glide	0–500 ms — portamento between notes
VR2	Sweep Time	1–2000 ms — duration of the one-shot pitch sweep (VR4 on LIVE)
VR3	Attack	0–1 s — fade-in time on each new trigger
VR4	Filter Env Amount	0–1 — how much the dedicated filter envelope opens the cutoff on attack
VR5	LFO Fade-In	0–2 s — time for the wail LFO to ramp up to full depth after a trigger
VR6	Chop Depth	0–1 — amplitude chopper depth
VR7	Chop Rate	0.5–8× — chopper rate, scaled to Wail Rate
VR8	LFO Phase Offset	0–1 — starting phase of the wail LFO on each trigger

7. Pot Pages — FX PAGE 4

Controls the onboard tape echo and reverb bus. The LIVE page's VR7/VR8 set how much signal reaches this bus; these eight knobs shape what happens to it once it's there.

Knob	Parameter	Range / Notes
VR1	Delay Time	20–1100 ms
VR2	Feedback	0–1.1 — above ~0.9 the echo self-oscillates into a howl
VR3	Tape Tone	800 Hz–10 kHz — low-pass filter in the echo feedback path (tape bandwidth)
VR4	Wow / Flutter	0–1% — pitch wobble depth on the echo repeats
VR5	Reverb Decay	0.70–0.95 — room size / tail length
VR6	Reverb Tone	1–18 kHz — damping (brighter = less damping)
VR7	Tape Drive	0–1 — saturation on the echo feedback signal
VR8	—	Unused, reserved for a future FX parameter

8. Line Inputs & the FX Bus

The LINE IN L/R jacks feed directly into the same tape echo and reverb bus as the siren. This means you can run another instrument or mixer send through DubSauce's echo/reverb while the siren plays over it — both sources mix into one bus and come out the stereo outputs together.

There's no separate send control for the line inputs — they always feed the bus at a fixed level. The FX page (Section 7) shapes the character for both the siren and anything on the line inputs equally.

> **Engine note:** the echo and reverb run entirely in the Seed3's internal RAM, capped at about 1.1 seconds of delay time. This was a deliberate design choice for rock-solid reliability on this hardware — see Section 11.

9. MIDI & CV

Signal	Behavior
MIDI Note On/Off	Gates the siren and sets its pitch for the held note's duration, overriding the Tune knob.
GATE IN 1	External trigger, equivalent to pressing and holding S1.
CV OUT 1	Outputs the siren's live wail/LFO value as a control voltage, for modulating external gear.

10. Quick Patch Recipes

Patch	How to get there
Classic air-raid wail	LIVE: Tune mid, Wail Rate ~2–4 Hz, Wail Depth high. SHAPE: Pulse wave.
Laser zap	LIVE: Sweep high, Decay short. DEEP: Sweep Time short (~50–150 ms).
Dub echo wash	LIVE: Delay Send and Reverb Send both up. FX: Feedback ~0.6–0.8, Tape Tone dark.
Self-oscillating dub	FX: Feedback past ~0.9. Let it run — the echo takes over on its own.
Smooth drone	SW18 down (latch). DEEP: Attack and LFO Fade-In both slow (0.5–1 s).

11. Known Limitations

- **Delay time is capped at ~1.1 seconds.** The echo buffer lives in the Seed3's internal RAM rather than external SDRAM, by design — on this board's boot configuration, external SDRAM isn't guaranteed to be ready before audio starts, so keeping the buffer internal is what makes the FX reliable. A longer-delay version using SDRAM is a possible future upgrade.
- **FX page knob 8 is currently unused** — reserved for a future parameter (candidates: ping-pong stereo spread, echo high-pass, or a second delay tap).
- **Resonance is capped at 0.9** to prevent filter self-destruction/silence at extreme settings.